



IDBI INN@VATE

2026

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Team Details

- a. **Team name:** SAARTHI
- b. **Team leader name:** Koushik Deb
- c. **Problem Statement:** Default Prediction Model

Brief about the idea

SAARTHI is an MSME loan default early-warning system. A lender uploads a loan dataset (CSV); SAARTHI auto-maps the columns, trains a calibrated model, and for every loan predicts the probability of default over the next 12 months.

But it never stops at a number. Each loan comes with a plain-English reason, a recommended action to reduce the risk, and a fairness check — and every explanation is verified by a second AI, so it cannot make things up.

In short: it turns a raw risk score into a decision a credit officer can act on.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How it is different: Most tools give only a risk score. SAARTHI adds a verified plain-English reason, a 12-month timeline, and a concrete fix for every loan — all in one common vocabulary, so any two loans are comparable.

How it solves the problem: The ML model gives a calibrated default probability; SHAP finds the drivers; an AI writes the reason and the action; a second 'faithfulness judge' checks it against the evidence; and a fairness audit guards against bias.

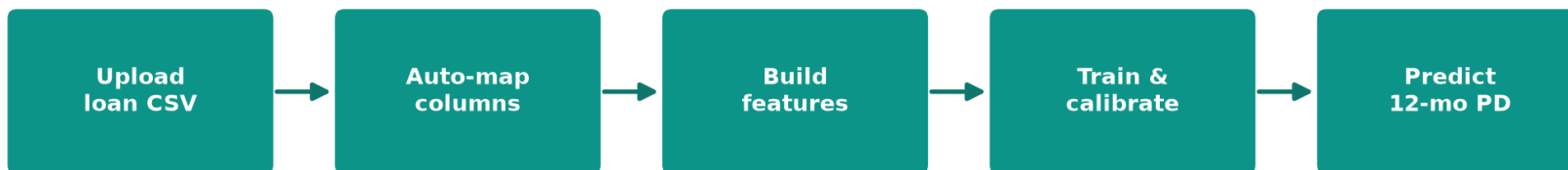
USP: Trustworthy, explained output — never a bare number. The AI can describe the model but is not allowed to invent it, and the single move that lowers the risk is shown as before to after.

List of features offered by the solution

- Auto column mapping — any CSV to a fixed schema, with user override
- Calibrated 12-month probability of default (PD) for every loan
- Early-warning risk curve that flags the onset month
- SHAP-based reasons in a fixed, comparable vocabulary (common interpretation framework)
- Plain-English explanation, verified by a faithfulness judge (anti-hallucination)
- Recommended action with before to after PD (recourse)
- Difference-aware fairness audit — protected attributes are never used to predict
- Portfolio dashboard + per-loan drill-down + model trace
- Multi-provider AI gateway with automatic fail-over; honest labelling of estimates

Process flow diagram or Use-case diagram

PHASE 1 — SCORE

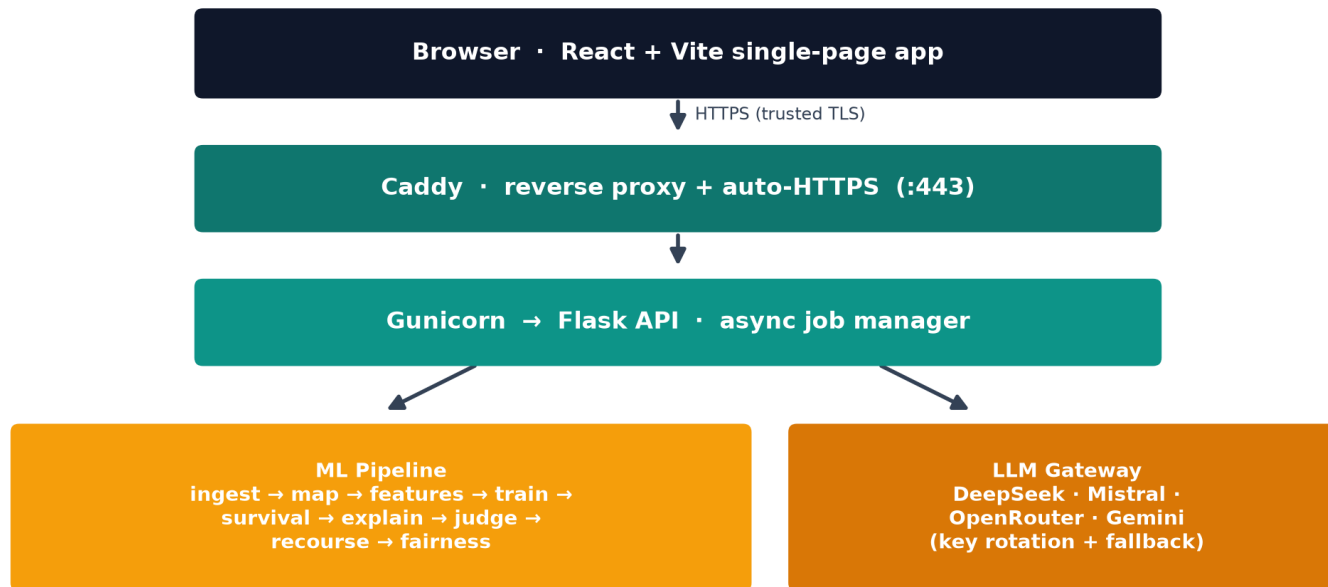


PHASE 2 — EXPLAIN & ACT



Wireframes/Mock diagrams of the proposed solution (optional)

Architecture diagram of the proposed solution



The ML model owns the number · the LLM only explains · a judge verifies · protected attributes are audit-only

Technologies to be used in the solution

Frontend: React, Vite, TypeScript, Tailwind CSS, Recharts

Backend: Python, Flask, Gunicorn (async job manager)

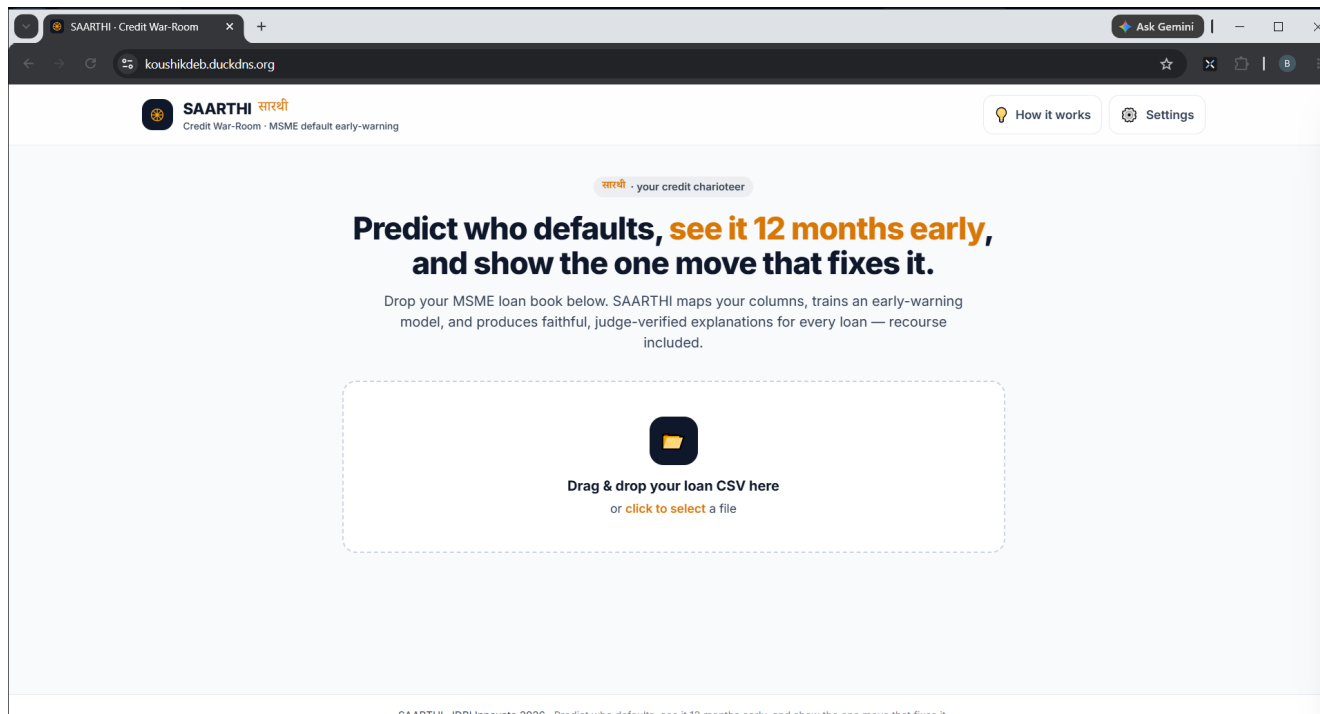
Machine Learning: LightGBM, scikit-learn, SHAP, lifelines (survival), fairlearn, pandas, polars

AI / LLM: DeepSeek, Mistral, OpenRouter, Google Gemini (OpenAI-compatible) · pydantic + json-repair

Hosting: Caddy (automatic HTTPS), systemd, DuckDNS, Ubuntu server

Estimated implementation cost (optional)

Snapshots of the prototype



Prototype Performance report/Benchmarking

- Accuracy (AUC-ROC): 0.97 on the real SBA loan book (~900K loans); 0.68–0.79 on German & MSME sets
- Calibration (ECE): about 0.05 — the probabilities are reliable, not just a ranking
- Explanation faithfulness: ~90–100% verified by the judge across test loans
- Scale & speed: scores 148K+ loans; trains in seconds; a per-loan explanation takes ~10–20s
- Fairness: demographic-parity & equalized-odds gaps reported per protected attribute — Pass on test data
- Reliability: 33/33 automated end-to-end checks and 12/12 live API checks pass

Additional Details/Future Development (if any)

- Integration with bank core / loan-origination systems and scheduled portfolio re-scoring
- Role-based access and on-premise / private deployment
- More models (TabPFN, XGBoost) and automatic data-drift monitoring
- Regulator-ready audit logs and exportable reason reports
- Research paper on the capability-diverse faithfulness judge

Provide links to your:

- GitHub Public Repository: https://github.com/DevDaring/IDBI_SAARTHI
- Demo Video Link (3 Minutes): <https://youtu.be/-HglTjzO5qc>
- Final Product Link: <https://koushikdeb.duckdns.org>



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Thank you!